CECS 451 Term Project, Team 10

LinkMate

A personal memory for your network that finds the right person for a problem, on your own laptop.

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Help exists in your network, but you can’t find it

The capture moment

You meet someone. What they do and offer ends up nowhere, or as a loose note.

The need moment

Weeks later you have a problem, and can’t remember who could help.

Contact books store a name and a company, not what a person can do. Acquaintances are a key source of new information and jobs (Granovetter 1973, Rajkumar et al. 2022), and their context is lost first.

Individuals who rely on people they barely know

Segment	Behavior	Role
Active networkers (primary)	Meet new people often, in many contexts: clubs, events, a new city	Early adopters, likely paying users
Occasional networkers (secondary)	Meet people less often, hit the same problem at specific moments	Mass market through free tier and word of mouth

Defined by behavior, not by job or age. Companies and sales teams are not targeted.

Write a note today, ask a question later

1 Capture

A note in your own words, or a pasted email

2 Review

AI proposes a profile, you correct and confirm

3 Ask

Describe a need in natural language

4 Act

Contacts with reason and source note, or “no suitable contact”

No forms, no tagging, and every recommendation is traceable.

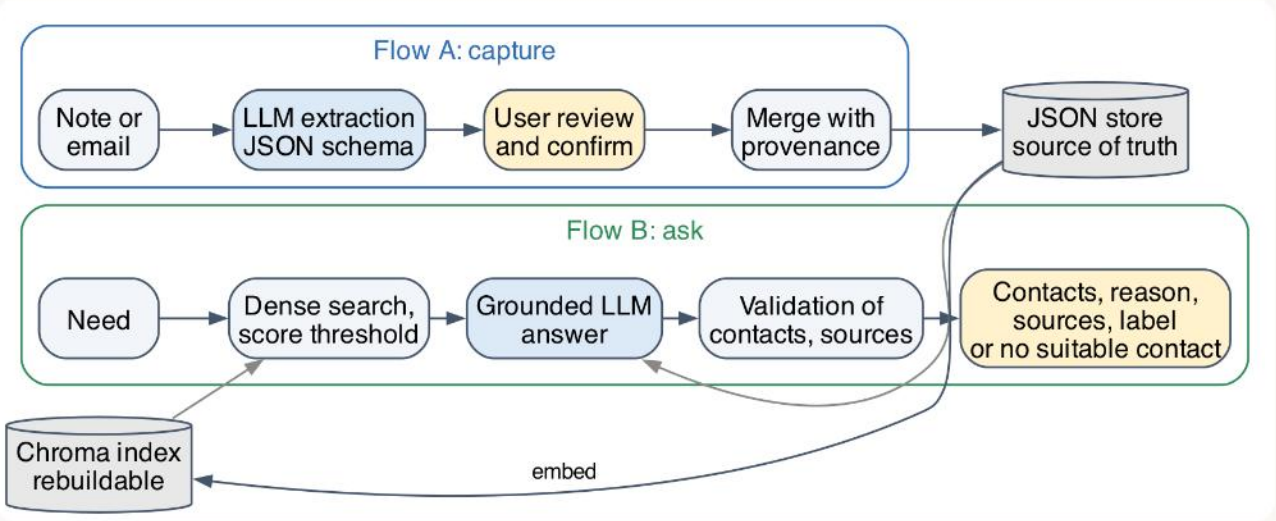
AI personal CRMs exist. The gap is narrower.

Product	Focus	Price
Dex	Personal CRM, notes, reminders	$12–20/month
Mesh	Network from connected accounts	Free, $10/month Pro
ConnectMachine	Digital cards, event capture	Free, $5.99/month
Happenstance	People search with explanations	Free, $24/month
Covve	Card scanning, enrichment	From $12/month

LinkMate: messy notes become enriched profiles, open-ended needs get a traceable source, and third-party data stays on the device.

Sources: vendor pricing pages, retrieved October 15, 2026

Local RAG, and the model only proposes



- Extraction with a fixed JSON schema
- Embeddings and cosine search
- Grounded answer from retrieved notes
- Code checks every source

qwen3:0.6b (0.5 GB) and nomic-embed-text through Ollama, on an 8 GB laptop, with no API keys. The larger qwen3:1.7b did worse.

Demo

The running prototype: note to profile, need to contact, and what happens when nobody fits.

Tested on questions it had never seen

55

synthetic contacts with 70 notes, some vague or very long

40

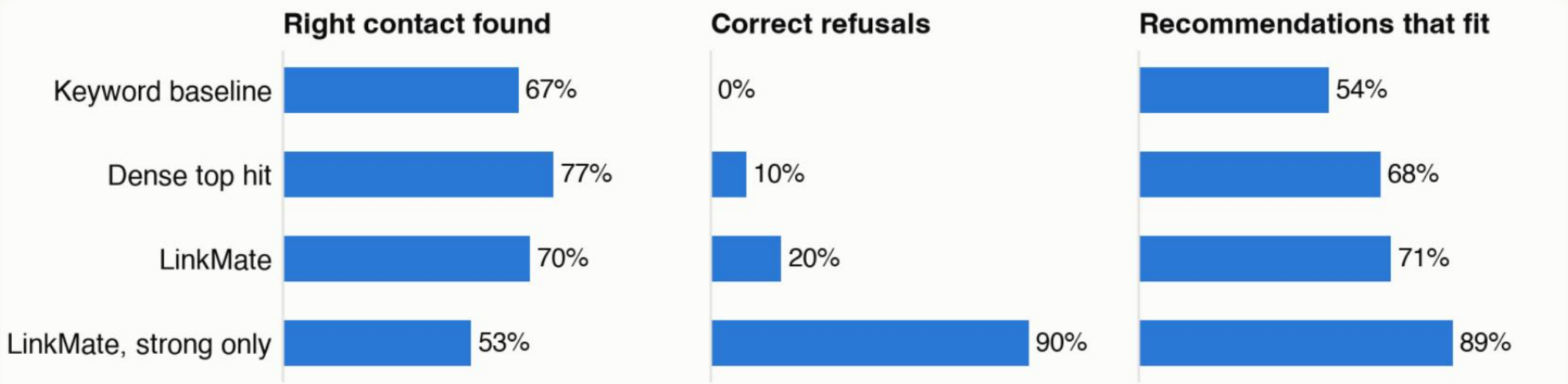
held-out questions, run after settings were frozen on 30 others

10

of them have no right answer, most with a near miss

Baseline: BM25 keyword search over the same notes. All numbers are reproducible from the repository.

Trust the strong label: 89% of those fit



Direct questions: 14/14. Paraphrased needs, found in the top three: 75% dense, 44% keywords. Overall LinkMate against keywords is not significant at this size (p = 0.51).

40 held-out questions: 30 answerable, 10 unanswerable. Weak recommendations fit 53% of the time.

Where it fails, and what still holds

Failures (17 of 40)

- Near misses accepted: a guitarist for Italian lessons
- Paraphrase too far from the note
- Reasons can invent details
- Two prompt fixes made it worse, so we dropped them

What held, and what to watch

- 0 invalid sources in all runs
- 70/70 notes extracted with the correct name
- Testing exposed an injection that steered two answers, now removed by a guard
- New wording still reaches the proposal, so review stays mandatory

Freemium, where inference is not the cost driver

Illustrative, not validated	Early	Reach
Users (0.1% of 64M US freelancers for reach)	10,000	64,000
Paying, at 4%	400	2,560
Revenue at $6/month	$2,400/month	$15,360/month
Inference if hosted, under 1 cent per user	$74/month	$474/month

Local models are the privacy argument, not the cost argument. A pilot for time saved and willingness to pay is prepared, not yet run.

Sources: Upwork 2023, First Page Sage via Crazy Egg (3–5% SaaS freemium), Curlscape pricing

Speed scales far past one person's network. Quality falls first.

Contacts	Full index	Search	Recall@3
55	4 s	44 ms	1.00
555	24 s	28 ms	1.00
2,055	87 s	25 ms	0.96
5,055	198 s	25 ms	0.79

Generated contacts, development questions. Above about 2,000 contacts, retrieval needs work, and reading all JSON files on each question should move to a database.

Clear limits, clear next steps

Limitations

- Weak refusals with the small model
- Synthetic data, no real users yet
- Wide confidence intervals at 40 questions
- Run on macOS only

Roadmap

1. User pilot with real notes, with consent
2. Reranker or stronger model for near misses
3. Optional encrypted sync for premium
4. Voice capture, imports, mobile capture view

The notes you already take become a network you can use.

Code, data and every number shown: github.com/VishalBahatyani/linkmate

Antoine Courant: business. Felix Baechle: integration, evaluation. Vishal Bahatyani: AI plan, repository, pilot. Mika Brueggemann: data, testing. [team confirms]